

## BREATH AS BIOLOGY, BREATH AS PRĀṆA



BREATH IS ONE OF THE MOST UNIVERSAL HUMAN EXPERIENCES, YET IT HOLDS TWO VERY DIFFERENT MEANINGS DEPENDING ON THE LENS WE USE. IN MODERN PHYSIOLOGY, BREATH IS UNDERSTOOD THROUGH RESPIRATORY MECHANICS, GAS EXCHANGE, LUNG CAPACITY, AND NERVOUS SYSTEM REGULATION. IN THE VEDIC-YOGIC TRADITION, BREATH IS PRĀṆA, THE LIFE-FORCE THAT ANIMATES BODY, MIND, AND CONSCIOUSNESS. THESE TWO APPROACHES ARE NOT CONTRADICTORY—THEY ILLUMINATE DIFFERENT ASPECTS OF THE SAME REALITY.

### VITAL CAPACITY: THE PHYSIOLOGICAL LENS

IN CONTEMPORARY SCIENCE, LUNG STRENGTH IS COMMONLY MEASURED USING VITAL CAPACITY, WHICH IS THE MAXIMUM AIR A PERSON CAN EXHALE AFTER A DEEP INHALATION. IN HEALTHY ADULTS THIS USUALLY RANGES BETWEEN 3 TO 5 LITERS, THOUGH AGE, HEIGHT, SEX, FITNESS, ALTITUDE, AND POLLUTION ALL INFLUENCE THE VALUES. RESEARCHERS ALSO USE THE VC INDEX, CALCULATED AS VITAL CAPACITY (ML) DIVIDED BY BODY WEIGHT (KG), TO COMPARE POPULATIONS. USING THIS METHOD, STUDIES CONSISTENTLY SHOW THAT WESTERN POPULATIONS TEND TO HAVE HIGHER AVERAGE VALUES, WHILE SOUTH ASIAN POPULATIONS OFTEN EXHIBIT SLIGHTLY LOWER VALUES EVEN WHEN MATCHED FOR HEIGHT AND AGE. THIS SUGGESTS A MIX OF GENETIC, STRUCTURAL, AND ENVIRONMENTAL FACTORS INFLUENCING LUNG PERFORMANCE.

### THE YOGIC LENS: BREATH AS PRĀṆA

CLASSICAL YOGA APPROACHES BREATH VERY DIFFERENTLY. THE HAṬHA-YOGA-PRADĪPIKĀ DEFINES BREATH THROUGH THREE PHASES:

PŪRAKA (INHALATION), KUMBHAKA (RETENTION), AND RECAKA (EXHALATION).

HERE, BREATH IS NOT SIMPLY A VOLUME OF AIR, BUT A MEDIUM FOR REGULATING PRĀṆA, THE SUBTLE LIFE-FORCE. THE GHERAṆḌA-SAMHITĀ INTRODUCES AN INTELLIGENT TIME-MEASUREMENT SYSTEM FOR BREATH CALLED MĀTRĀ. A MĀTRĀ ROUGHLY EQUALS THE DURATION OF ONE CALM HEARTBEAT. INSTEAD OF COUNTING SECONDS, YOGIS COUNTED BREATH IN MĀTRĀS. IN MODERN TERMS, ONE MĀTRĀ MAY BE APPROXIMATELY 0.5 TO 1 SECOND, DEPENDING ON THE PRACTITIONER'S STATE. SO IF A YOGI HELD KUMBHAKA FOR 80 MĀTRĀS, IT WOULD CORRESPOND TO APPROXIMATELY 40 TO 80 SECONDS, DEPENDING ON HEART RATE AND TRAINING LEVEL. THE CHĀNDOGYA UPANIṢAD TAKES THIS FURTHER, DESCRIBING PRĀṆA AS THE FUNDAMENTAL SUPPORT OF LIFE. THIS SHIFTS BREATH FROM A MECHANICAL FUNCTION TO A SPIRITUAL PRINCIPLE.

### NEUROMUSCULAR EFFECTS OF PRĀṆĀYĀMA

MODERN RESEARCH NOW CONFIRMS EFFECTS THAT YOGIS OBSERVED EXPERIENTIALLY. SLOW, CONTROLLED BREATHING:

- STRENGTHENS THE DIAPHRAGM AND INTERCOSTAL MUSCLES
- STIMULATES THE VAGUS NERVE
- INCREASES PARASYMPATHETIC TONE
- IMPROVES RIB-CAGE MOBILITY
- INCREASES CO<sub>2</sub> TOLERANCE DURING KUMBHAKA



# MINDFUL SHAPES YOGA



## BREATH AS BIOLOGY, BREATH AS PRĀṆA



THESE ADAPTATIONS INFLUENCE BOTH PHYSIOLOGY AND EMOTIONAL REGULATION. STRESS HORMONES REDUCE, HEART RATE DECREASES, BLOOD PRESSURE STABILIZES, AND MENTAL FLUCTUATIONS BEGIN TO SETTLE. IN THIS WAY, PRĀṆĀYĀMA BECOMES BOTH A PHYSICAL AND PSYCHOLOGICAL PRACTICE.

### TWO VIEWS, ONE REALITY

PHYSIOLOGY EXPLAINS HOW BREATH SUPPORTS LIFE THROUGH OXYGEN TRANSPORT, RESPIRATORY MECHANICS, AND NEURAL PATHWAYS. YOGA EXPLAINS HOW BREATH SUPPORTS LIFE THROUGH PRĀṆA, MENTAL STILLNESS, AND EXPANSION OF AWARENESS. ONE SPEAKS IN LITERS AND SECONDS, THE OTHER IN MĀTRĀS AND CONSCIOUSNESS. TOGETHER, THEY OFFER A MUCH RICHER UNDERSTANDING OF WHAT IT MEANS TO BREATHE.

### CONCLUSION

WHEN WE BREATHE UNCONSCIOUSLY, BREATH SIMPLY MAINTAINS SURVIVAL. WHEN WE BREATHE CONSCIOUSLY, BREATH BECOMES A TOOL FOR HEALING, CLARITY, AND INNER TRANSFORMATION. SCIENCE HELPS US UNDERSTAND BREATH AS BIOLOGY. YOGA HELPS US EXPERIENCE BREATH AS PRĀṆA. BOTH VIEWS ARE NEEDED. BOTH ARE TRUE.

